

Ross Hoopchuk

+1-501-581-7196 | rhoopchuk3@gatech.edu | th00p.com | Atlanta, GA
 [thoopchuk26](#) |  [thoopchuk](#)

SUMMARY

Master of Computer Science candidate specializing in Computer Graphics with strong expertise in C++ programming, 3D graphics rendering, and software development. Experience in full-stack web development, game engine architecture, and low-level systems programming. Proven ability to develop complex technical projects including custom graphics engines and emulators.

EDUCATION

- Georgia Institute of Technology** 08/2024 - 05/2026
Master of Science in Computer Science, Specialization: Computer Graphics
Atlanta, GA
 - GPA: 3.5/4.0
 - Relevant Coursework: Computer Graphics, Computer Vision, Machine Learning, Advanced Algorithms
- Hendrix College** 08/2019 - 05/2023
Bachelor of Science in Computer Science
Conway, AR
 - GPA: 3.4/4.0
 - Capstone Thesis: [Custom 3D Graphics Engine in C++](#)

TECHNICAL EXPERIENCE

- Hendrix College Solar Eclipse Website** 08/2022 - 12/2022
Full-Stack Web Developer
Conway, AR
 - Developed full-stack web application for managing 2024 Solar Eclipse event with 500+ expected attendees
 - Designed and implemented SQL database schema for volunteer and booth registration systems
 - Built responsive front-end interfaces using .NET framework with C# and JavaScript
 - Deployed application on Microsoft Azure cloud platform with continuous integration
 - Collaborated in agile team environment with version control using Git

TECHNICAL PROJECTS

- 3D Graphics Rendering Engine** 03/2023 - 05/2023
C++, Windows API, Graphics Programming, Linear Algebra 
 - Engineered real-time 3D rasterizer from scratch implementing lighting models, vertex/pixel shaders, and OBJ file parsing
 - Developed camera systems, transformation matrices, and clipping algorithms for 3D model visualization
 - Optimized rendering pipeline achieving 60 FPS for complex scenes with multiple light sources
- Game Boy Emulator** 05/2024 - 06/2024
C++, Low-Level Programming, CPU Architecture, Graphics Rendering 
 - Implemented accurate Game Boy CPU emulation supporting 500+ opcodes with cycle-accurate timing
 - Developed PPU (Picture Processing Unit) graphics renderer with sprite priority and background scrolling
 - Created memory management unit with bank switching for ROM cartridge support
 - Built debugger with disassembler and memory viewer for development and testing
- Zero-Knowledge Proof Identity Verification System** 04/2025 - 05/2025
Rust, Cryptography, Mobile Development, Security 
 - Developed privacy-preserving age verification system using ZKCredits library for zero-knowledge proofs
 - Implemented secure data extraction from digital passports to mobile application
 - Created verifier service that validates proofs without accessing sensitive user information
 - Applied cryptographic principles for secure identity verification in compliance scenarios

TECHNICAL SKILLS

- Programming Languages:** C++ (Advanced), Python (Intermediate), JavaScript (Intermediate), Rust (Intermediate), C# (Proficient), SQL (Proficient)
- Graphics & Game Development:** OpenGL, DirectX, 3D Graphics Rendering, Shader Programming, Game Engine Architecture, Computer Vision
- Web Development:** .NET Framework, ASP.NET, JavaScript, HTML/CSS, REST APIs, Azure Cloud Services
- Databases:** MySQL, SQL Server, Database Design, Query Optimization
- Development Tools:** Git, Visual Studio, Jupyter Notebooks, NumPy, Pandas, Docker, CI/CD
- Areas of Expertise:** Computer Graphics, Systems Programming, Emulation Development, Cryptography, Full-Stack Development, Software Architecture

ADDITIONAL INFORMATION

Languages: English (Native), Mandarin (Intermediate - B1 Level)
Certifications: Georgia Tech Graduate Student in Computer Science
Interests: Real-time Graphics Programming, Game Engine Development, Computer Vision Applications, Cryptography Systems, Open Source Contributions